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Outcomes for posterior uveal melanoma: Validation of American Brachytherapy Society Guidelines

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ABSTRACT

PURPOSE: To assess outcomes of small and medium choroidal melanoma (less than 5.0 mm in height) following Iodine-125 episcleral brachytherapy.

METHODS AND MATERIALS: Patients with small and medium choroidal melanoma that underwent Iodine-125 brachytherapy with apical height of 1.0 mm to 5.0 mm and largest basal diameter of ≤16.0 mm were included. Data were extracted from the original dosimetry plans to determine doses to vision critical structures with the prescription point to the apical height (actual dose, ABS guidelines) and, after simulation, with the prescription point to the height of 5.0 mm (simulated dose, COMS protocol). Visual acuity (VA) outcomes with actual dose and that predicted with the simulated dose were estimated along with local recurrence, ocular survival, and survival at 5 years.

RESULTS: A total of 339 patients with a mean age of 61.5 years with a mean follow up duration of 43.4 months were included. The mean dose reduction for lens, optic disc, and fovea was 34%, 39.4%, and 41.4%, respectively with actual dose when compared with simulated dose. The Kaplan-Meier estimations for 3 year event free rate of VA of 20/50 or better were 56% and 31% for actual dose and simulated dose, respectively. Only 3 events of local recurrence were observed (enucleated) yielding 5 year local control and ocular survival rate of 98%. Overall survival (OS) and metastasis free survival (MFS) were 95% and 87.5% at 5 years, respectively.

CONCLUSIONS: Small and medium choroidal melanoma treated according to ABS has excellent outcomes. Brachytherapy planning using ABS guidelines as compared to COMS protocol may be associated with lower rates of radiation toxicity and vision loss.

Keywords: ABS guidelines; COMS protocol; Choroidal melanoma; Brachytherapy; Iodine-125; Vision loss

Introduction

Brachytherapy is the most frequent primary treatment for uveal melanoma in the United States (1). A recent survey has identified a variation in the brachytherapy planning of which therapeutic dose and prescription point are most relevant (2). Therapeutic dose for brachytherapy using Iodine-125 of 85 Gy as prescribed in the Collaborative Ocular Melanoma Study (COMS) has become the standard (3). However, the prescription point was variable depending upon the tumor height; for tumors more than 5.0 mm in height, the prescription point was the actual tumor height, whereas for tumors less than 5.0 mm in height, the prescription point was adjusted to 5.0 mm. In 2012, the American Brachytherapy Society (ABS) (TG-129 report) issued a recommendation of using either actual tumor height as the prescription point or 5.0 mm for tumors less than 5 mm (4). The guidelines were revised by ABS in 2014 (Level 1 consensus) specifying that the prescription point should be the actual tumor height for all tumors including those that were less than 5.0 in height (5). The panel cited concerns

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Table 1
Choroidal melanomas up to 5.0 mm in tumor height: extracted demographics and outcome measures from literature review

Authors	Isotope	*N* Total (tumors <2.5 mm)	Prescription point	Recurrence rate	Ocular survival	Metastasis	Visual Acuity*		Mean Follow-up (Years)
							20/40 - 20/60	<20/200	
Finger et al, 1999[a]	Pd-103	80 (12)	5.0 mm	2%	94%	3%	83%	8%	7.0
Damato et al, 2005[b]	Ru-106	458 (126)	5.0 mm	2%	99%	N/R	N/R	N/R	9.0
Verschueren et al, 2010[c]	Ru-106	425 (84)	5.0 mm	3%	N/R	N/R	60%	N/R	5.0
Berry et al, 2013[d]	I-125	82 (0)	5.0 mm	2%	98%	11%	35%	43%	3.9
Murray et al, 2013[e]	I-125	55	Apical Height	2%	4%	7%	52%	31%	12.5
		40	5.0 mm	0%	3%	10%	65%	30%	
Sanchez-Tabernero, 2017[f]	I-125	311 (28)	Apical Height	5%	N/R	N/R	N/R	N/R	6.5
Jiang et al, 2020[g]	Ru-106	39 (9)	5.0 mm	7%	100%	4%	37%	33%	5.8
Pagliara et al, 2020[h]	Ru-106	152	Apical Height	N/R	N/R	N/R	54%	78%	5.6
Present study, 2021	I-125	339 (154)	Apical Height	1%	98%	4%	44%	21%	3.6

N/R = not reported.

*Visual acuity outcomes divided in 2 groups. This shows the percentage of patients that maintained a VA of between 20/40 and 20/60, and those that had a VA worse than 20/200.

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about dose toxicity as one of the main factors for their recommendation.

We have recently demonstrated that by using current ABS guidelines, excellent outcomes can be achieved for small choroidal melanoma treated with Iodine-125 brachytherapy. The majority (69%) of patients retained visual acuity of 20/50 or better with very high local control and ocular survival rate (99.3%) with absence of metastasis (100%) (6). In the present study, we have expanded the spectrum of treated tumors to include all tumors that are subject to prescription point guidelines (height of 5.0 mm) and include most of the T1 stage tumors of the AJCC tumor size criteria (7). We treated these tumors using ABS guidelines and simulated the outcomes for the COMS protocol.

Although several studies have reported outcomes of uveal melanoma treated with brachytherapy with Iodine-125 and other radionuclides (8–18), careful review reveals only 2 studies that display data for tumors up to 5.0 mm in apical height treated with Iodine 125 prescribed to apical height (ABS guidelines) and report functional (visual) outcomes (14, 15) and survival rates (overall and metastasis free survival rate)(Table 1) (14). As of the writing of this study, we have no knowledge of any studies that have reported dosimetric comparisons between doses delivered using prescription point of tumor height (actual dose, ABS guidelines) and to the height of 5.0 mm (simulated doses, COMS protocol) with quantification of dose to vision critical structures (lens, optic disc, fovea) and associated functional (visual) outcomes using model based vision

Table 2
Choroidal melanoma: study demographics and treatment planning data by apical height

Factor	1.0–2.5 mm $N = 154$ Mean ± SD Value (%)	2.6–5.0 mm $N = 185$ Mean ± SD Value (%)	Total $N = 339$ Mean ± SD Value (%)
Eye			
R	80 (52.2)	102 (55.1)	182 (53.7)
L	74 (47.8)	83 (44.9)	157 (46.3)
Age (years)	59.3 ± 14.2	63.4 ± 13.9	61.5 ± 14.0
Gender			
Female	60 (39.0)	103 (56.8)	163 (48.1)
Male	94 (61.0)	82 (43.2)	176 (51.9)
Mean follow up (months)	40.5 (37.2)	40.1 (31.4)	43.4 (35.0)
Apical height (mm)	2.1 ± 0.3	3.8 ± 0.7	3.0 ± 1.04
Largest basal diameter (mm)	8.2 ± 2.1	10.3 ± 2.4	9.7 ± 2.6
Distance to fovea (mm)	4.6 ± 4.8	5.8 ± 5.0	4.0 [1.03, 7.7]
Distance to optic disc (mm)	5.2 ± 4.2	6.3 ± 4.3	4.5 [3.0, 7.5]
Prior or adjuvant therapy	0	0	0
Dose to apex (Gy)	85.4 ± 11.2	86.5 ± 5.1	87.0 ± 9.3
Status			
Alive	148 (96.1)	160 (86.5)	308 (90.8)
Dead due to metastasis	0 (0)	6 (3.2)	6 (1.8)
Dead due to other causes	6 (3.9)	19 (10.3)	25 (7.4)

Table 3
Dose in Gy to vision critical structures and reduction by prescription point at apical height (actual)

Structure	At apical height (actual) Mean (SD)	At 5.0 mm (simulated) Mean (SD)	Dose reduction (mean percent)
Lens	10.61 (9.99)	16.07 (13.57)	34%
Optic Disc	28.74 (20.12)	47.40 (34.32)	39.40%
Fovea	57.26 (50.86)	97.66 (92.19)	41.40%

Table 4
Number of patients who experienced threshold dose (> 45 Gy) to vision critical structures by prescription point at apical height (actual)

Structure	At apical height (actual)	At 5.0 mm (simulated)	Threshold exposure (reduction)
Lens	6	10	40%
Optic Disc	55	147	63%
Fovea	150	197	24%

loss prediction. Additional outcomes including local recurrence, ocular survival, and survival (overall and metastasis free survival rate) were also analyzed.

Methods and materials

Demographics

Patients treated for choroidal melanoma at the Cole Eye Institute, Cleveland Clinic, Ohio between January 2004 and December 2017 were included in the initial clinical chart institutional review board (IRB) - approved review. An exempt status from the IRB at the Cleveland Clinic Foundation was obtained to collect and analyze anonymous patient information. The inclusion criteria for this study were small and medium tumors per COMS criteria (apical height of 1.0 mm to 5.0 mm and large basal diameter (LBD) less than 16.0 mm) (3). Patients that received any form of prior therapy or adjuvant transpupillary thermotherapy (TTT) were excluded. Patient basic demographics and their clinical information were obtained from the electronic medical record system (EpicCare EMR, Epic Systems Corp) and paper medical records.

Brachytherapy planning

The patients received a comprehensive initial work up at the time of presentation. They underwent a detailed ophthalmoscopic examination, photography, and ultrasonographic A and B scans. Tumor size (height and basal diameter) was determined by ultrasound and ophthalmoscopic examination. Accurate plaque placement before the introduction of intraoperative ultrasonographic confirmation in June 2006 was assessed by use of indirect ophthalmoscope or transillumination. Patients were followed using standard clinical protocol of eye examinations every 3 months for

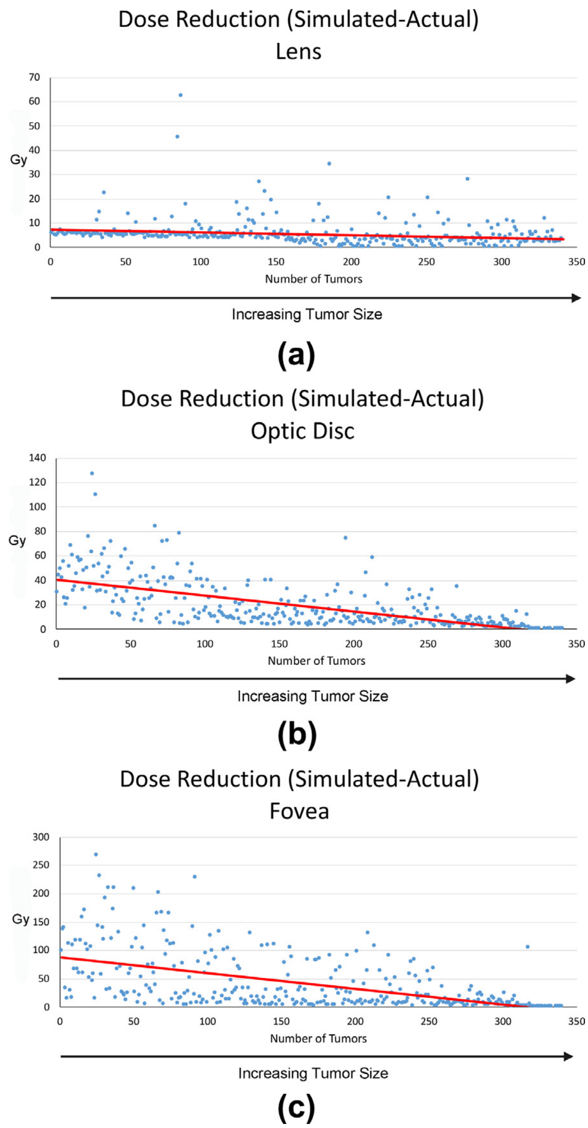

Fig. 1. Reduction in radiation exposure by tumor height. The magnitude of reduction in dose exposure (simulated dose (COMS)- actual dose) to lens (A), optic disc (B), and fovea (C) decreased with increasing tumor height.

the first year, every 6 months for up to 5 years, and annually thereafter.

Commercially available treatment planning software (Plaque Simulator V.5.3.4; Bebig GmbH, Berlin, Germany) was used for determining dosimetry for the brachytherapy implants. Plans were based on retinal diagrams showing the size (radial, circumferential, and height) and position (distances from fovea and disc) of the target. We used the dose-volume histogram to assist in choosing the best plaque between COMS 10 to COMS 20, COMS notched (All COMS plaques manufactured by Trachsel Dental Studio, Inc., Rochester, NY), and Eye Physics 917 (Manufactured by Eye Physics, LLC, Los Alamitos, CA). Iodine-125 was the radionuclide used in the study with apical dose prescription of 85 Gy, assuming the eye and adjoining tissue are composed of water. For the 5.0 mm prescription height simulated cases, the original plan was modified as to prescription point, the source activity reassigned to deliver 85 Gy to 5.0 mm rather than tumor apex, using the same treatment time. The dose was recalculated and doses to points of interest were extracted.

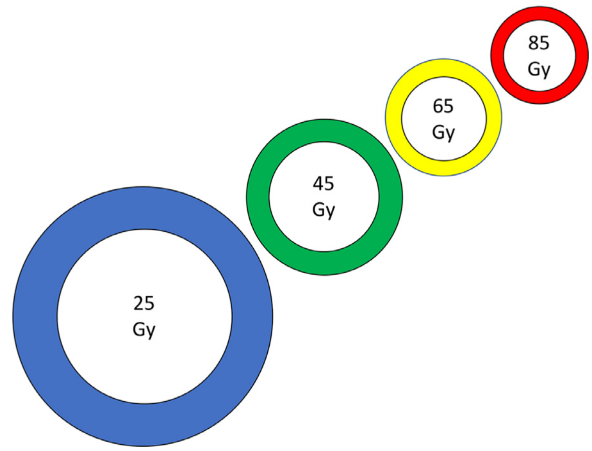

Fig. 2. Proportional representation of mean retinal surface area exposed to 25, 45, 65, and 85 Gy. The mean retinal surface exposed with the actual dose (prescription point, apical height) is represented by a white circle. The colored band is depicting the excess of retinal surface exposed with simulated dose (prescription point, 5.0 mm).

Outcomes

A. Doses to vision critical structures

Doses to vision critical structures of the eye defined as lens, optic disc, and fovea were extracted from the original dosimetry plans that specified the prescription point to the actual apical height of the tumors (actual dose) and to the height of 5.0 mm (simulated dose) and the percent dose reduction to each vision critical structure was calculated. Threshold dose values were set at 45 Gy for lens, optic disc, and fovea and a scatter plot was generated for easier visualization. Retinal surface area (mm^2) that received dose of at least 25Gy, 45 Gy, 65, and 85 Gy was also calculated for actual dose and simulated dose.

B. Functional (vision)

Best corrected visual acuity was recorded using a Snellen eye chart measured at every visit and was converted into the logarithm of the minimum angle of resolution values for statistical analysis. Baseline visual acuity was first divided 20/50 or better and 20/200 or better, respectively. A mathematical model was then developed to estimate vision loss based on mean actual dose levels. To develop this model, Cox proportional hazards models were fit based on actual dose to predict VA loss at 20/50 and 20/200, separately. Using these models, the average actual doses for each vision critical structure were calculated and survival curves for vision loss based on these doses were estimated. Then, average doses to vision critical structures

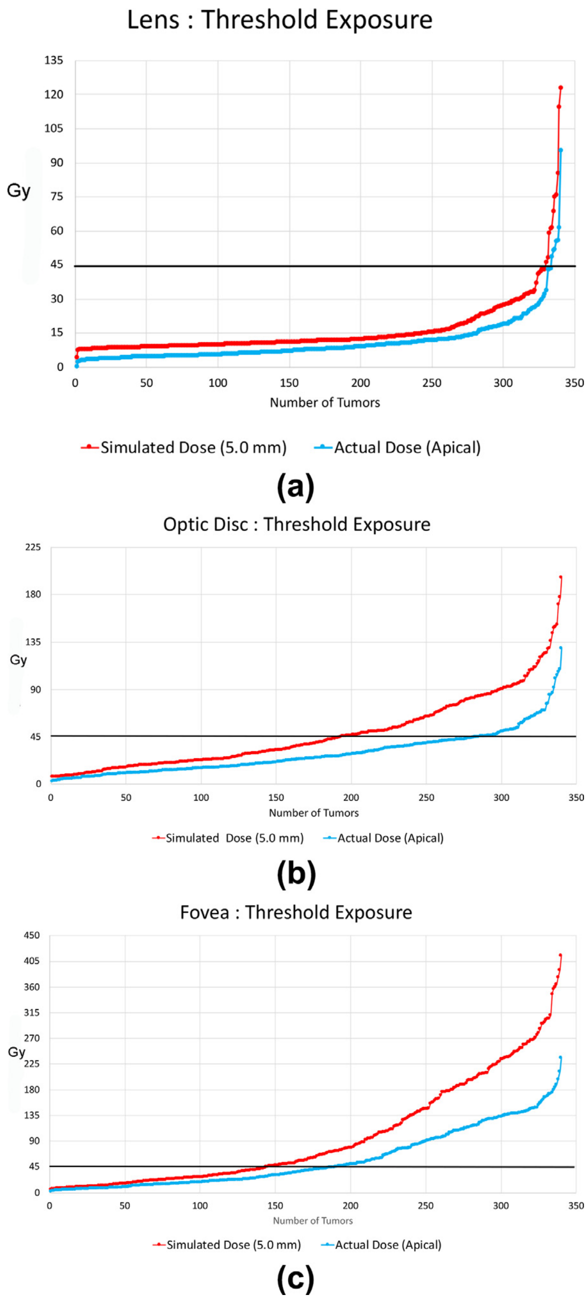

Fig. 3. Threshold exposure (45 Gy) to vision critical structures lens (**A**), optic disc (**B**), and fovea (**C**) with actual dose (prescription point, apical height: orange) and for simulated dose (prescription point, 5.0 mm, blue). In all cases, lens, optic disc, and fovea were exposed to lower radiation dose with fewer tumors receiving suprathreshold (>45 Gy) exposure with actual dose (prescription point, apical height) than for simulated dose (prescription point, 5.0 mm).

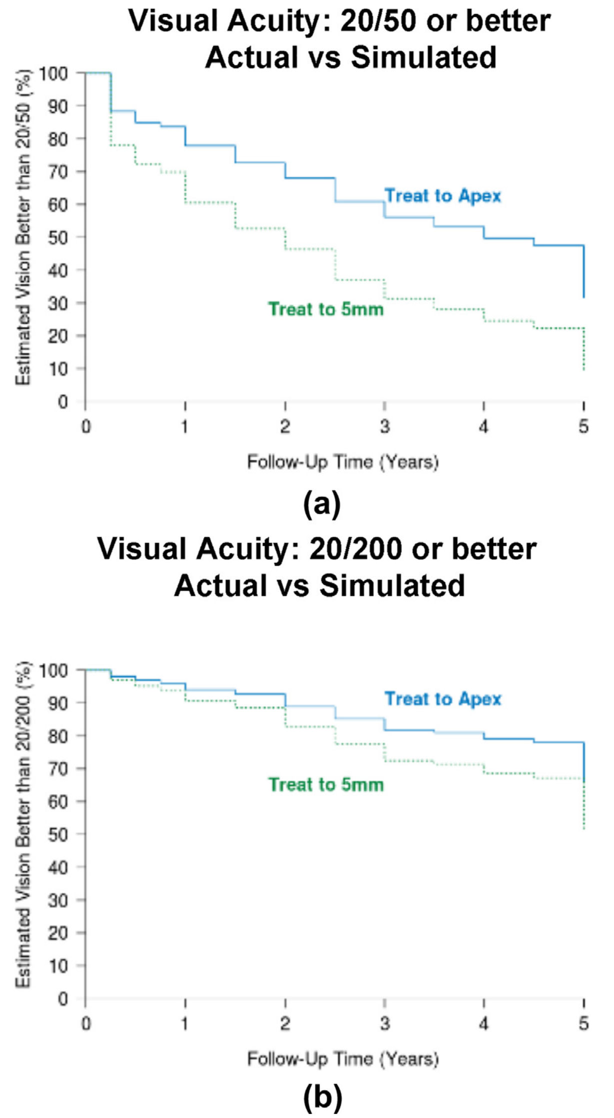

Fig. 4. Estimates of event-free rate of VA of 20/50 or better (**A**) and 20/200 or better (**B**) for mean actual dose (prescription point, apical height) and predicted mean simulated dose (prescription point, 5.0 mm).

using simulated dose were estimated, and the vision losses based on these estimated doses were calculated using the survival curves estimated from the actual dose data. Of note, the average dose estimates from the simulated data all fell within the observed actual dose ranges, so no extrapolation beyond the range of the original model was needed.

C. Local control and ocular survival

Local recurrence was determined by tumor growth of >0.2 mm in apical height and >0.5 mm in basal diameter compared to pretreatment tumor dimensions (19). Kaplan-Meier estimates were performed to estimated recurrence at 1, 3, and 5 years. Ocular survival was calculated as the absence of enucleation.

D. Survival

Metastasis was excluded at baseline by systemic staging (CT scan of chest, abdomen, and pelvis) followed by surveillance using hepatic ultrasonography every 6 months for the first 5 years and annually thereafter. Survival outcomes were reported as overall survival (OS) and metastasis free survival (MFS) rate using Kaplan-Meier estimates at 1, 3, and 5 years.

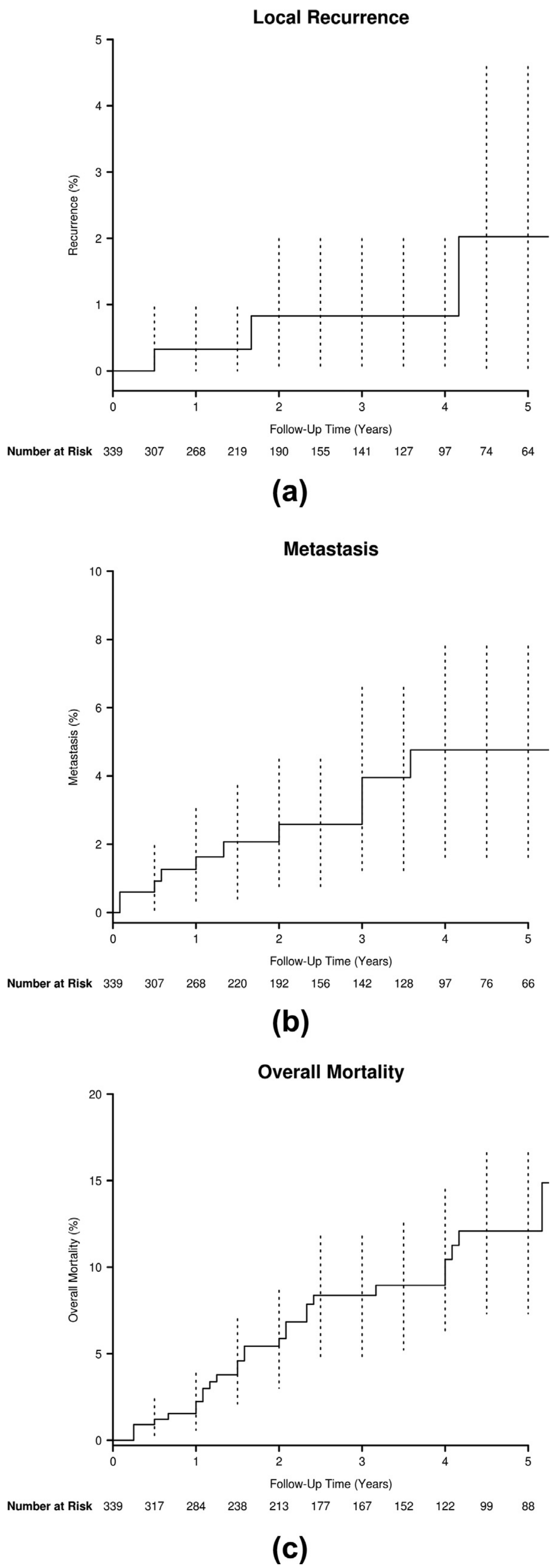

Fig. 5. Kaplan-Meier estimates (5 year) of local recurrence (**A**) metastasis-free survival (**B**), and overall survival (**C**).

Results

Demographics

A total of 339 patients treated with iodine-125 brachytherapy between 1 January 2004 and 31 December 2017 were included in this study. The mean (standard deviation, SD) age of the included patients was 61.5 (14.0) years at the start of treatment. One hundred seventy six (52%) were male. The mean (SD) apical height for the tumors was 3.0 (1.04) mm and the mean (SD) large basal diameter was 9.7 (2.6) mm. The mean (SD, median) follow up was 43.4 (35, 37) months. The patients were divided into small and medium tumors (COMS criteria). Data on small choroidal melanoma patients (161) from our previous study were included except for 7 patients whose dosimetry planning data could not be located (Table 2).

Outcomes

A. Doses to vision critical structures

The mean lens dose was 10.61Gy and 16.07 Gy for actual dose and simulated dose, respectively with a mean dose reduction with actual dose of 34%. The optic disc mean dose was 28.74 Gy and 47.40 Gy for actual dose and simulated dose, respectively with a mean dose reduction with actual dose of 39.4%. The mean fovea dose was 57.26 Gy and 97.66 Gy for actual dose and simulated dose, respectively with a mean dose reduction with actual dose of 41.4% (Table 3). The magnitude of dose reduction decreased with increasing tumor height with no difference observed between the actual dose and simulated dose for tumors of height 5.0 mm (Fig. 1). Doses to vision critical structures expressed as threshold dose (45 Gy) for lens, optic disc, and fovea were reduced in 40%, 63%, and 24% of eyes, respectively for actual dose compared with simulated dose (Table 4, Fig. 2). The reduction in retinal surface (area mm^2) receiving at least 25 Gy, 45 Gy, 65 Gy, and 85 Gy was 31%, 27%, 27%, and 29% respectively for actual dose compared with simulated dose (Table 5, Fig. 3).

B. Functional (vision)

Best corrected visual acuity distribution at baseline showed that 74% (252) of patients had a VA of 20/50 or better and 98% (332) had a VA of 20/200 or better. Using mean actual dosing levels the estimated 3 year event free rates for VA of 20/50 or better and VA of 20/200 or better were 56% and 82%, respectively. When mean estimated doses for treating to 5 mm were used, the 3-year event-free rates for VA of 20/50 or better and VA of 20/200 or better were 31% and 73%, respectively (Fig. 4).

C. Local control and ocular survival

Only 3 events of local recurrence following treatment were observed in the study (confirmed by ultrasonography and fundus photographs), with a 5 year event free rate

Table 5
Dose in Gy to retinal surface (area mm^2) at varying levels and reduction achieved by prescription point at apical height (actual)

Retinal exposure (simulated) Gy	At apical height (actual) Surface area dose (reduction) Mean (SD)	At 5.0 mm Mean (SD)	Percent
Retina 25	26.98 (9.85)	39.23 (8.39)	31%
Retina 45	17.25 (6.24)	23.54 (6.00)	27%
Retina 65	13.12 (4.89)	18.07 (4.50)	27%
Retina 85	10.56 (4.20)	14.90 (3.72)	29%

of 98% (Fig. 5). All 3 eyes with recurrent tumors were enucleated yielding a 5 year ocular survival rate of 98%.

D. Survival

Twelve of the study patients developed metastasis. Six patients died from metastasis related complications. Twenty five patients in this cohort died of unrelated causes. OS and MFS were 95% and 87.5% at 5 years, respectively (Fig. 5).

Discussion

All cases included in this study were treated using a single radionuclide (Iodine-125) with planned apical height dose of 85 Gy as per ABS guidelines (5). The spectrum of treated tumors was subject to prescription point guidelines (height of 5.0 mm) that also encompassed most of the T1 stage tumors of the AJCC tumor size criteria (7). To minimize selection bias, the study cohort is of a consecutive case series based upon clinical diagnosis rather than cytologic or histologic confirmation as per the current standard of care. All patients were treated at the Department of Ophthalmic Oncology of the Cole Eye Institute and were followed using a standardized protocol (19).

We treated these tumors using ABS guidelines. As per COMS protocol, all tumors in this study would have been treated to the prescription height of 5.0 mm regardless of actual apical height in contrast to ABS guidelines that recommend prescription point of actual tumor height. Since increased tumor size is an important predictor of vision loss, it can be assumed that a higher dose delivered to vison critical structures using COMS protocol would have led to a higher risk of vision loss (20). In fact, the ABS panel cited concerns about dose toxicity as one of the main factors for their recommendation. To our knowledge, we are not aware of any studies that have reported dosimetric comparisons between doses delivered using prescription point of tumor height (actual dose, ABS guidelines) and to a height of 5.0 mm (simulated doses, COMS protocol) with quantification of dose to vision critical structures (lens, optic disc, fovea). In the presence of conflicting practice patterns, the superiority of one recommendation over the other cannot be resolved readily without a balanced randomized clinical trial. Given constraints of resources, lack of funding for orphan diseases, and rarity of uveal melanoma, we compare observed visual outcome (actual dose, ABS) versus predicted visual outcome (simulated dose, COMS) using a dose-driven mathematical model developed to estimated vision loss for simulated dose (described above in methods section). While these outcomes are only estimates, the mean dosing levels used to estimate vision loss were within the range of observed dosing levels, and do not reflect extrapolation outside the range of observed data.

Absolute dose to each vision critical structure (lens, optic disc, fovea) was reduced by 34%, 39.4 %, and 41.4 %, respectively with actual dose (ABS) when compared with simulated dose (COMS) (Table 3**).** When expressed as threshold dose (45 Gy), 40%, 63%, and 24% of eyes were spared with actual dose compared with simulated dose (Fig. 3). However, as expected, the benefit of dose sparing to vision critical structures with actual dose (ABS) when compared with simulated dose (COMS) reduced with increasing tumor height (Fig. 1). The reduction in dose was reflected in the predicted visual gain (VA of 20/50 or better) in 25% of patients as the Kaplan-Meier estimations for a 3 year event free rate were 56% and 31%, respectively. The visual gain was less (9%) for worse visual acuity (20/200 or better) (Fig. 4). Until effective therapies for radiation retinopathy are developed (21), avoiding it by minimizing dose to the retina seems reasonable.

Only 3 local recurrences were observed in the study (managed by enucleation) with a 5-year event-free rate and 5 year ocular survival rate of 98% (Fig. 5). Such high rates of local control and ocular survival have also been reported with alternate radionuclides used for brachytherapy such as Ru-106 (2%-7%) (12, 13, 17), Pd-103 (2%), (9), and proton beam therapy (5%) (22). The majority of recurrences following brachytherapy are evident in the first 3 years and the mean (SD, median) follow up of the present study cohort of 43.4 (35, 37 months) seems to be adequate, it is certainly possible that local recurrence rates may be higher with longer follow up (15, 19, 23).

Our results expand on observations in a small number of cases that were not enrolled in COMS and were treated using ABS guidelines; the authors did not notice any differences in visual acuity, local recurrence, and ocular survival. Moreover, the excellent local outcomes in the present study mirror outcomes reported with small choroidal melanoma. More importantly, the low local recurrence rate observed when tumors were treated with apical height dose of 85 Gy (ABS guidelines) alleviates the need for a higher dose to be delivered when treating to a height of 5.0 mm (COMS protocol) (3, 24, 25). In fact, radiation planning using COMS protocol may be associated with worse visual outcomes.

Metastasis was excluded at baseline by systemic staging (CT scan of chest, abdomen, and pelvis) followed by surveillance using hepatic ultrasonography every 6 months for the first 5 years and annually thereafter (26). High OS and MFS (95% and 87.5% at 5 years, respectively) seem to reflect small and medium tumor size (COMS) in this study (Fig. 5). Our survival results are supported by the previous COMS report (27), AJCC TNM cohort study (28) as well retrospective studies (9, 17). Despite growing evidence that local control may not be curative in uveal melanoma (29), local control is the cornerstone of the management of uveal melanoma as local recurrence may be a marker for a higher risk of metastasis (30).

The limitations of this study include its retrospective nature and limited follow up duration. The bias inherent in any retrospective was minimized by using standard protocol based assessment for both ocular examination and systemic surveillance (26, 19). Considering the rarity of uveal melanoma, the study sample is acceptable. However, to achieve a more definitive assessment of low frequency events such as recurrence and metastasis reported herein, a larger number of samples would be desirable. The follow up duration also seems to be adequate (mean 43.4 months, median 37 months) given that the median time to recurrence following brachytherapy is 18 months with the majority (70%) of the recurrences occurring within the first 3 years (19).

Conclusions

Small and medium choroidal melanomas treated according to ABS guidelines have excellent local outcomes. Brachytherapy planning following ABS guidelines as compared to COMS protocol guidelines may be associated with lower rates of radiation toxicity and vision loss.

Contributorship statement

KM: Data collection, critical revision of manuscript.

RJY: Data collection, critical revision of manuscript.

ZR: Data collection, interpretation, drafting, critical revision of manuscript.

RJFB: Data analysis and interpretation, critical revision of manuscript.

AW: Study design, data acquisition, data interpretation, drafting manuscript.

JS: Data interpretation, drafting of manuscript, and interpretation.

AS: Study design, data interpretation, drafting manuscript, critical revision of manuscript.

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